DELHI URBAN HEAT INTELLIGENCE

A GIS, Remote Sensing and Machine Learning Based Analysis of Urban Heat Patterns in Delhi

PROJECT REPORT

Submitted by: Shruti Verma

Department: Computer Science & Engineering

Institution: Sister Nivedita University

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This project, titled "Delhi Urban Heat Intelligence: A GIS, Remote Sensing and Machine Learning Based Analysis of Urban Heat Patterns in Delhi," was planned and completed by me.
I acknowledge the freely available data and tools that made this study possible: the Landsat 8 and Landsat 9 imagery provided by the USGS and NASA, the Google Earth Engine platform, the FAO GAUL administrative boundaries, QGIS, and the open-source Python libraries used for the analysis.

ABSTRACT

This project analyses urban heat patterns in Delhi using Geographic Information Systems (GIS), remote sensing and machine learning. Landsat 8 and Landsat 9 Collection 2 Level-2 imagery for April to June 2023 was processed in Google Earth Engine to derive the Normalized Difference Vegetation Index (NDVI), Land Surface Temperature (LST) and an Urban Heat Island (UHI) layer at 30 m resolution.

The rasters were exported as float32 GeoTIFF files and converted into a dataset of 1,673,718 valid pixels using Python. The mean surface temperature of Delhi was 41.62 °C, and NDVI and LST showed a negative correlation of −0.54, which confirms that vegetation helps reduce surface heat. A Random Forest model that predicts LST from NDVI achieved an R^2 of 0.324 and a mean absolute error of 1.73 °C. The predicted temperatures were grouped into four heat categories (Low, Moderate, High and Extreme). QGIS was used for visualization and map preparation.

Keywords: GIS, Remote Sensing, NDVI, Urban Heat Island, LST, Landsat, Google Earth Engine, QGIS, Python, Machine Learning, Random Forest, Delhi.

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1. INTRODUCTION

Urban Heat Island (UHI) is a phenomenon in which urban areas experience higher temperatures than their surroundings. Rapid urbanization, dense buildings, roads and reduced vegetation all contribute to this effect.

This project analyses urban heat patterns in Delhi using GIS, remote sensing and machine learning. Satellite imagery is processed in Google Earth Engine to calculate NDVI, Land Surface Temperature (LST) and a UHI layer. The results are then analysed statistically and modelled in Python, and the maps are visualized in QGIS.

The study helps to understand the relationship between vegetation, urban development and temperature distribution in Delhi.

2. OBJECTIVES OF THE STUDY

The main objectives of this project are:

1. To analyse vegetation distribution in Delhi using NDVI.
2. To study urban heat patterns using UHI data.
3. To analyse Land Surface Temperature (LST).
4. To measure the relationship between vegetation and surface temperature.
5. To build a machine learning model that predicts LST from NDVI.
6. To classify the study area into heat categories.
7. To use Google Earth Engine, QGIS and Python for mapping and spatial analysis.

3. STUDY AREA

The study area selected for this project is Delhi, India. The boundary of Delhi was taken from the FAO GAUL 2015 administrative boundaries dataset (level 1).

Delhi is a highly urbanized region with residential areas, commercial centres, roads, industries and green spaces. Due to rapid urbanization, the region shows strong variations in surface temperature and vegetation cover. The study focuses on understanding these spatial variations using satellite imagery.

4. DATA SOURCES AND TOOLS

The project uses Landsat 8 and Landsat 9 Collection 2 Level-2 satellite imagery. The satellite data was processed for the period 1 April 2023 to 30 June 2023, using images with cloud cover below 20 percent. A total of 14 images met these conditions.

The study uses three main parameters:

- **NDVI:** Normalized Difference Vegetation Index.
- **LST:** Land Surface Temperature, in degrees Celsius.

- **UHI:** Urban Heat Island, calculated as the difference between LST and the mean LST of Delhi.

The following software and tools were used:

- **Google Earth Engine:** image filtering, cloud masking, calculation of NDVI, LST and UHI, statistics and export of rasters.
- **QGIS 3.44:** raster visualization and preparation of the final maps.
- **Python 3:** data extraction and modelling using pandas, NumPy, rasterio, scikit-learn, matplotlib, seaborn and joblib.
- **Visual Studio Code:** development environment for the Python scripts.
- **Microsoft Word:** preparation of the project report.

5. METHODOLOGY

The project was carried out in the following stages.

5.1 Data Collection

The Landsat 8 and Landsat 9 Collection 2 Level-2 image collections were merged and filtered by the Delhi boundary, the date range (April to June 2023) and cloud cover below 20 percent.

5.2 Cloud Masking

The QA_PIXEL band was used to remove fill pixels, dilated clouds, cirrus, clouds, cloud shadows and snow from every image. This prevents cloudy pixels from producing false low temperatures and wrong vegetation values. A median composite of the masked images was then created and clipped to the Delhi boundary.

5.3 NDVI Calculation

Surface reflectance bands were first scaled to reflectance values. NDVI was then calculated from the near infrared (band 5) and red (band 4) bands:

$$Reflectance = DN \times 0.0000275 - 0.2$$

$$NDVI = (NIR - Red) / (NIR + Red)$$

5.4 Land Surface Temperature (LST)

LST was calculated from the surface temperature band (ST_B10) and converted to degrees Celsius:

$$LST\ (°C) = ST_B10 \times 0.00341802 + 149.0 - 273.15$$

5.5 Urban Heat Island (UHI)

The UHI layer shows how much each pixel is warmer or cooler than the average temperature of Delhi:

$$UHI = LST - Mean\ LST\ of\ Delhi$$

Hotspots were defined as pixels where LST is higher than the mean LST plus one standard deviation.

5.6 Raster Export and Dataset Creation

NDVI, LST and UHI were exported from Google Earth Engine as single-band float32 GeoTIFF files at 30 m resolution in the EPSG:32643 (UTM zone 43N) coordinate system. The files were exported as raw values and not as styled images, so that each pixel keeps its real NDVI or temperature value.

A Python script read the three rasters, checked that they share the same grid (1755 × 1653 pixels), removed pixels outside the Delhi boundary and invalid pixels, and saved the result as a table of 1,673,718 valid pixels.

5.7 Machine Learning Model

Two regression models, Random Forest and Linear Regression, were trained to predict LST (target) from NDVI (feature). A random sample of 300,000 pixels was used, split into 80 percent training data and 20 percent test data. The Random Forest used 100 trees with a minimum of 20 samples per leaf. Model quality was measured with the Mean Absolute Error (MAE) and the R^2 score.

The UHI layer was deliberately not used as a feature. UHI is LST minus a constant, so using it would give the model the answer directly and produce a misleading R^2 of 1.0.

5.8 Heat Categories

The temperatures predicted by the Random Forest model were grouped into four categories: Low (below 40 °C), Moderate (40 to 42 °C), High (42 to 44 °C) and Extreme (above 44 °C).

6. RESULTS AND ANALYSIS

6.1 Thematic Maps

Three thematic maps were generated using Google Earth Engine and displayed in QGIS:

- **NDVI Map:** shows the distribution of vegetation across Delhi.
- **LST Map:** shows spatial variations in land surface temperature.
- **UHI Map:** shows how much warmer or cooler each area is compared with the Delhi average.

In the UHI map, the Yamuna river corridor and other vegetated or water areas appear cooler, while the western and south-western parts of Delhi show the strongest warm areas.

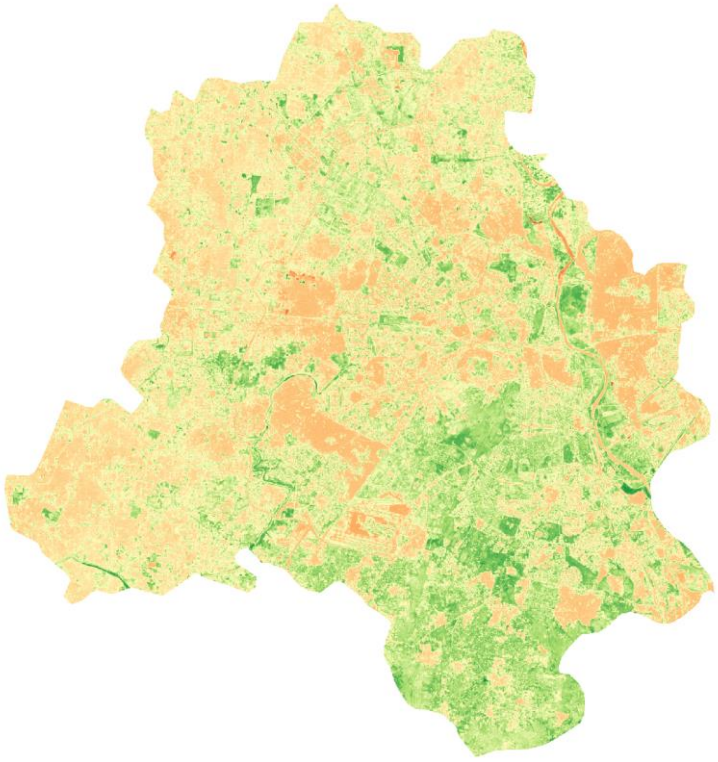


Figure 1: NDVI Map of Delhi

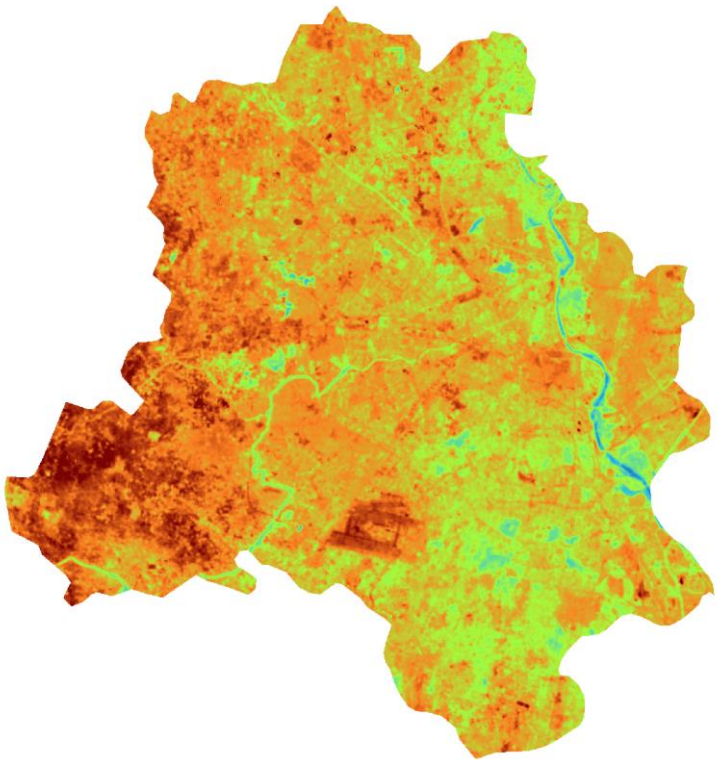


Figure 2: Land Surface Temperature (LST) Map of Delhi

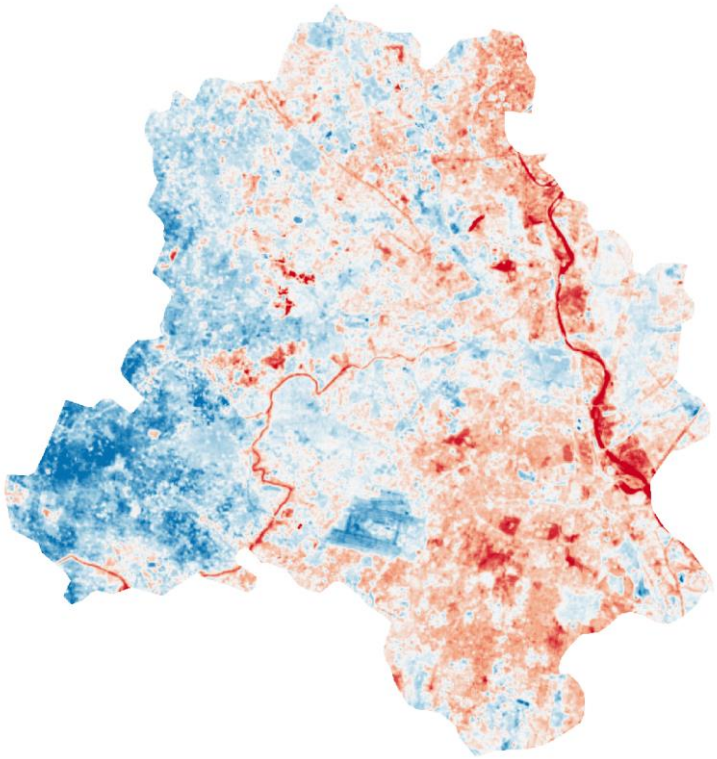


Figure 3: Urban Heat Island (UHI) Map of Delhi

6.2 Statistical Summary

Table 1 summarizes the processed data. The values were checked against the statistics calculated in Google Earth Engine, and the mean LST is identical in both (41.62 °C).

Table 1: Summary of the processed data

Parameter	Value	Unit / note
Landsat images used	14	Landsat 8 and 9
Study period	April–June 2023	cloud cover < 20%
Spatial resolution	30	metres
Valid pixels	1,673,718	after cleaning
Mean LST	41.62	°C
Standard deviation of LST	2.79	°C
Minimum LST	29.05	°C
Maximum LST	53.74	°C
Hotspot threshold (mean + 1 SD)	44.41	°C
NDVI range	−0.16 to 0.83	mean 0.32
UHI range	−12.57 to 12.12	°C

6.3 Relationship between NDVI and LST

The Pearson correlation between NDVI and LST was −0.54. The negative sign means that pixels with more vegetation tend to have lower surface temperature, which supports the role of green cover in reducing urban heat. A correlation of this size also means that vegetation explains only part of the temperature variation, because built-up surfaces, materials and building density also affect LST.

The correlation between UHI and NDVI is identical (−0.54), as expected, because UHI differs from LST only by a constant value.

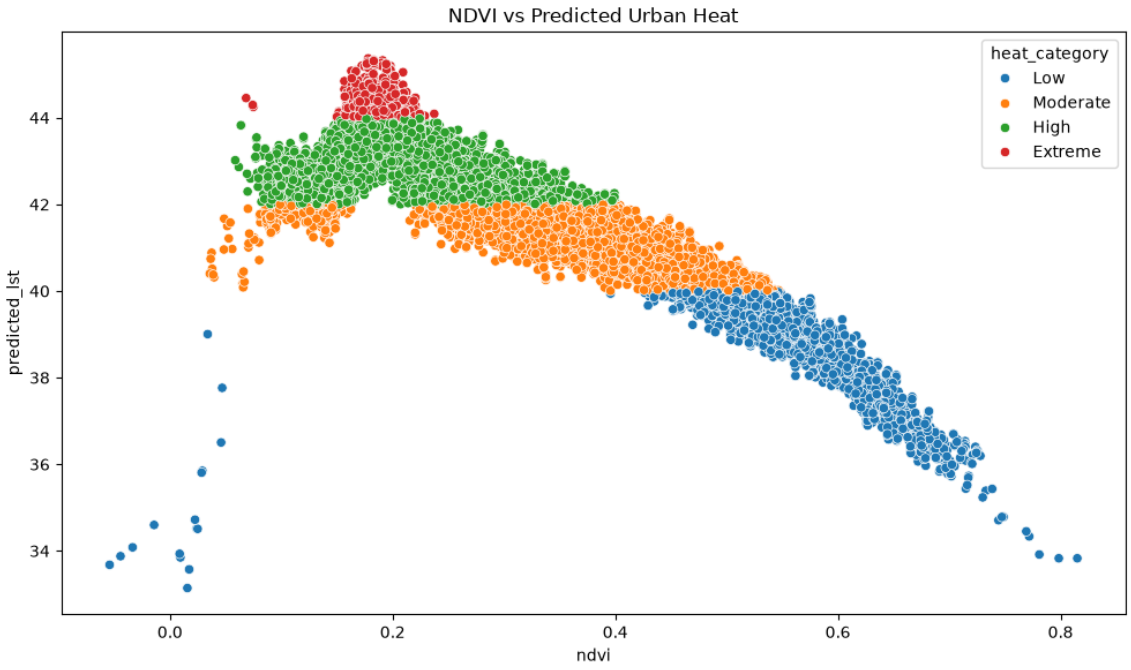


Figure 4: NDVI versus predicted urban heat

6.4 Model Performance

Table 2: Model performance on the test data

Model	MAE (°C)	R^2 score
Random Forest	1.73	0.324
Linear Regression	1.76	0.297

The Random Forest predicts LST with an average error of about 1.7 °C and explains about 32 percent of the variation in temperature using NDVI alone. This agrees with the correlation analysis, because the square of −0.54 is about 0.29. Random Forest performed slightly better than Linear Regression.

6.5 Predicted Temperature and Heat Categories

The predicted temperatures ranged from 33.14 °C to 45.38 °C, with a mean of 41.62 °C. Table 3 shows how the pixels were distributed among the four heat categories.

Table 3: Heat category distribution

Category	Predicted LST	Pixels	Share
Low	below 40 °C	289,086	17.3%
Moderate	40–42 °C	576,396	34.4%
High	42–44 °C	730,075	43.6%
Extreme	above 44 °C	78,161	4.7%
Total		1,673,718	100%

About 48 percent of the pixels fall in the High or Extreme categories, which shows that a large part of Delhi experiences very high surface temperatures during April to June.

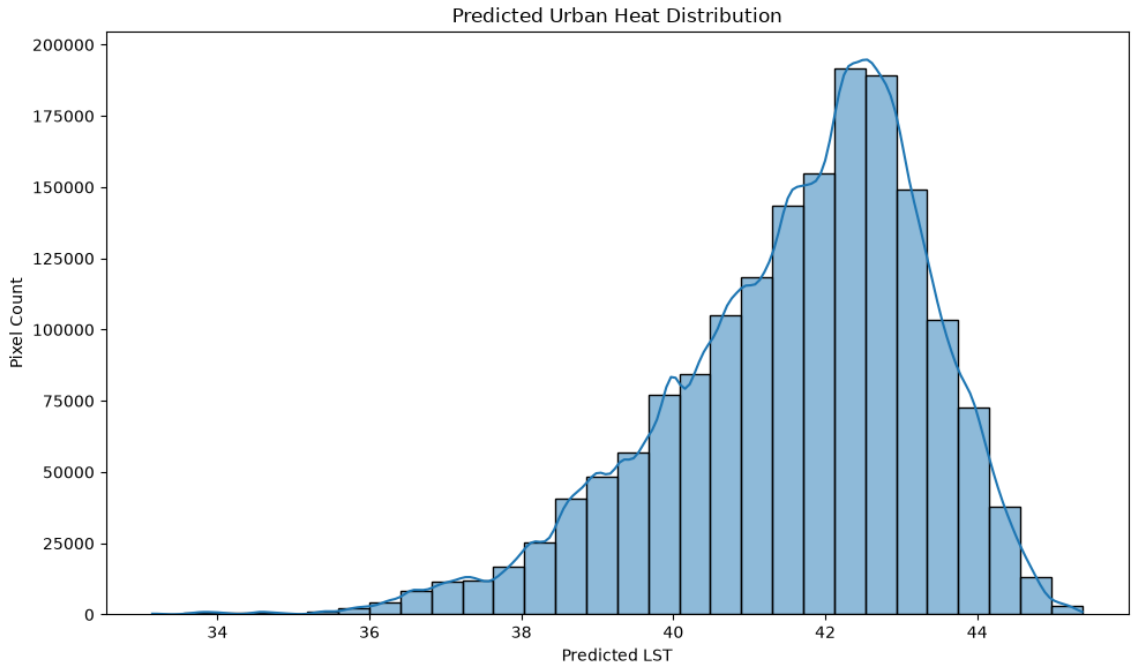


Figure 5: Predicted urban heat distribution

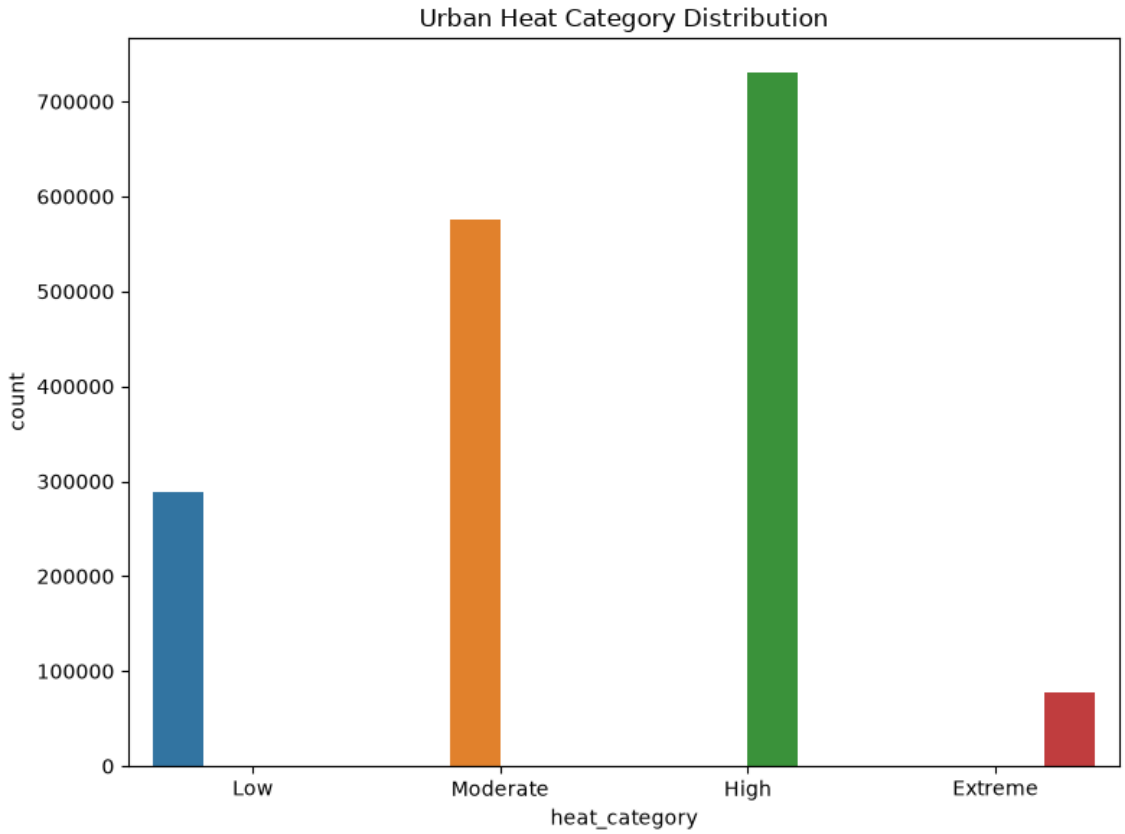


Figure 6: Urban heat category distribution

7. LIMITATIONS

- The model uses NDVI as its only feature, so it explains only about 32 percent of the variation in LST.
- Predictions are smoother than the real satellite temperatures (33.1–45.4 °C predicted compared with 29.0–53.7 °C observed), so the heat categories mainly show the effect of vegetation, and the most extreme observed temperatures are under-represented.
- The analysis uses a single median composite for April to June 2023, so it does not show changes between years or seasons.
- Landsat thermal data has a native resolution of 100 m and is resampled to 30 m, so very small hot or cool areas may not be captured.

8. CONCLUSION

This project demonstrates the use of GIS, remote sensing and machine learning for studying urban heat patterns in Delhi. NDVI, LST and UHI layers were generated from Landsat 8 and 9 imagery using Google Earth Engine, analysed in Python and visualized in QGIS.

The mean surface temperature of Delhi was 41.62 °C, and the negative correlation of −0.54 between NDVI and LST shows that vegetation helps to cool the urban surface. A Random Forest model predicted LST from NDVI with an R^2 of 0.324 and a mean error of 1.73 °C, and

the study area was divided into four heat categories, with about 48 percent of the pixels in the High and Extreme categories.

The project highlights the importance of vegetation monitoring and geospatial technology in urban environmental planning. The results can be improved in future work by adding more predictors, such as a built-up index (NDBI), elevation and land use data, and by analysing several years of imagery.

9. REFERENCES

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